

HOMEWORK 3

1. Prove that the ring $\mathbb{Z}[\sqrt{2}]$ is Euclidean.
2. Find a non-principal ideal in $\mathbb{Z}[\sqrt{-5}]$.
3. Prove that if p is a prime integer such that $p \equiv 3 \pmod{4}$, then p is a prime element of $\mathbb{Z}[i]$. Is 2 prime in $\mathbb{Z}[i]$?
4. Let p be a prime integer with $p \equiv 1 \pmod{4}$. Prove that $p = a^2 + b^2$ for some integers a and b . (Hint: Use the ring $\mathbb{Z}[i]$ of Gauss numbers.)
5. Let n be an integer that has at least two distinct presentations as the sum of two squares $a^2 + b^2$ of positive integers a and b . Prove that n is not a prime integer. (Example: $1000009 = 1000^2 + 3^2 = 972^2 + 235^2 = 293 \cdot 3413$ is a product of two primes!)
6. Let a and b be nonzero elements of a UFD R . Prove that the ideal $\gcd(a, b)$ contains both ideals (a) and (b) and it is contained in every principal ideal that contains both ideals (a) and (b) .
7. Prove that if $R[x]$ is a UFD, then so is R .
8. Using Eisenstein's Criterion prove that the polynomial $x^9 + y^9 + z^9$ is irreducible in $\mathbb{C}[x, y, z]$.
9. Show that over any field there exist infinitely many monic irreducible polynomials.
10. Prove that the polynomial ring $\mathbb{Q}[x_1, x_2, x_3, \dots]$ in infinitely many variables is not Noetherian but a UFD.